16. Slide Verification Problem #16

Slope, homogenous, water table

16.1. Introduction

This model is taken from Arai and Tagyo (1985) example #3, and it consists of a simple slope of homogeneous soil with pore pressure.

16.2. Problem Description

Verification problem #16 is shown in Figure 16.1. The material properties are given in Table 16.1. The location for the water table is shown in Figure 16.1. The position of the critical slip surface and the corresponding factor of safety is calculated for both a circular and noncircular slip surface. Pore pressures are calculated assuming hydrostatic conditions. The pore pressure at any point below the water table is calculated by measuring the vertical distance to the water table and multiplying by the unit weight of water. There is zero pore pressure above the water table.

16.3. Geometry and Material Properties

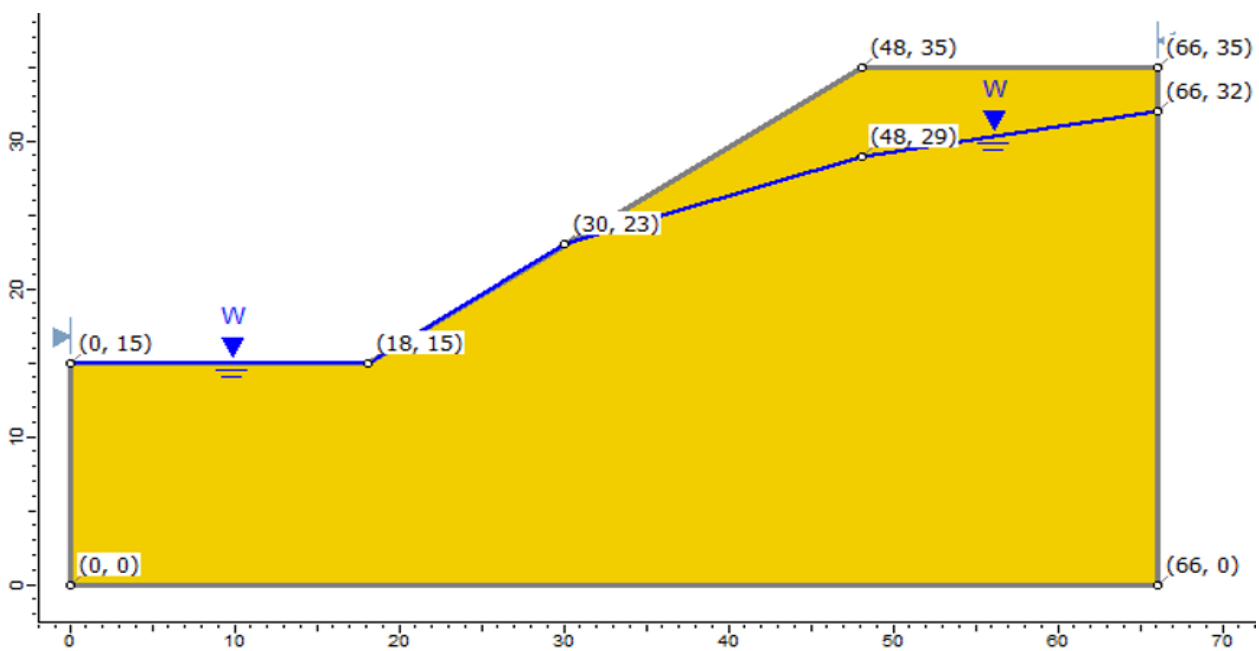

Figure 16.1: Geometry Setup in **Slide**

Table 16.1: Soil Properties

	c´ (kN/m^2)	ϕ´ (deg.)	γ (kN/m^3)
Soil	41.65	15	18.82

16.4. Results

Table 16.2: Circular – using auto refine search

Method	Factor of Safety
Bishop	1.118
Janbu Simplified	1.046
Janbu Corrected	1.131
Spencer	1.118

Note: Arai and Tagyo (1985) Bishops Simplified factor of safety = 1.138

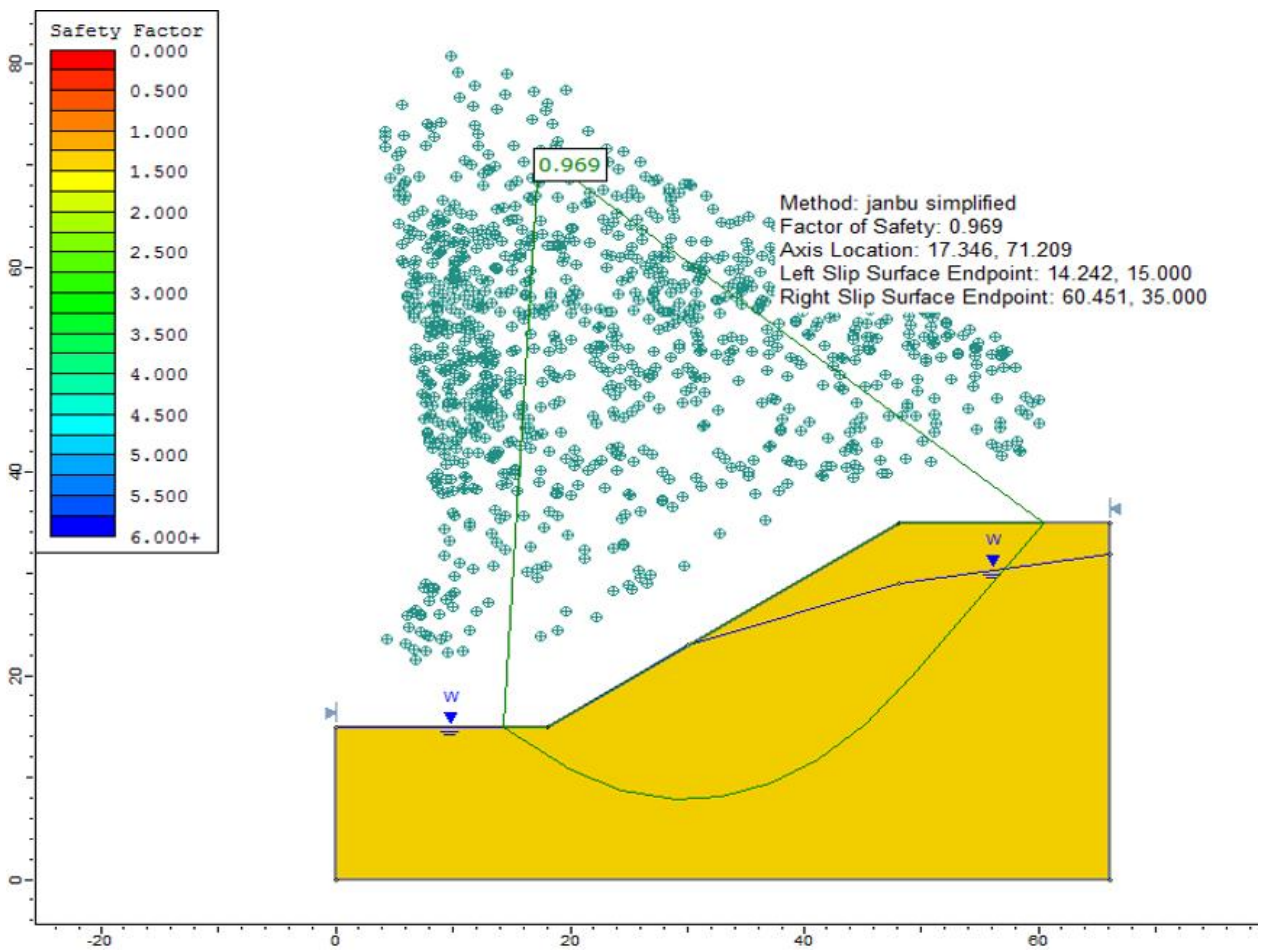

Figure 16.2: Circular failure surface using the Bishop simplified method

Table 16.3: Noncircular – using Random search with Monte-Carlo optimization

Method	Factor of Safety
Janbu Simplified	0.968
Janbu Corrected	1.050
Spencer	1.094

Note: Arai and Tagyo (1985) Janbu Simplified Factor of Safety = 0.995

Arai and Tagyo (1985) Janbu Corrected Factor of Safety = 1.071

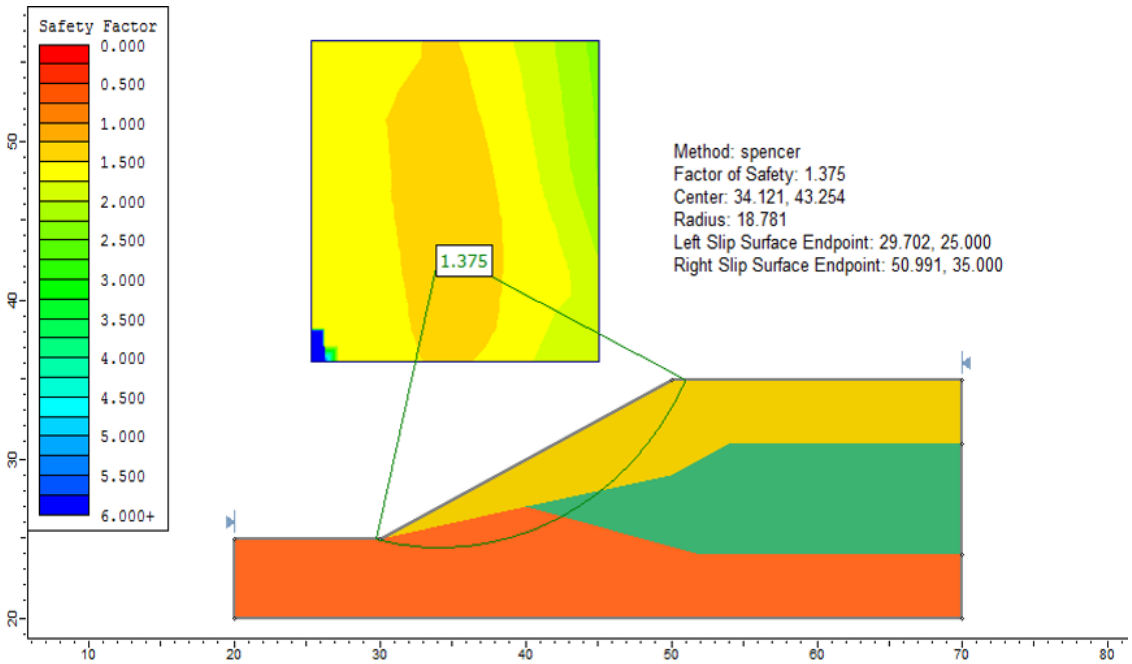

Figure 16.3: Noncircular failure surface using Janbu simplified method